

Assignment No 7

Aim: Develop a Java application that simulates an ATM machine. Implement functionalities like checking account balance, withdrawing, and depositing money. Use try, catch, and finally blocks to handle potential exceptions such as insufficient funds (throwing `ArithmeticException`) and invalid input (throwing `IllegalArgumentException`). Ensure that the application continues to run smoothly after handling exceptions.

Code:

```
import java.util.InputMismatchException;
import java.util.Scanner;

class ATM_Machine{
    private double balance;

    ATM_Machine(double balance) {
        this.balance = balance;
    }

    public void checkBalance() {
        System.out.println("Current Balance: ₹" + balance);
    }

    public void deposit(double amount) {
        if (amount <= 0) {
            throw new IllegalArgumentException("Deposit amount must be positive");
        }
        balance += amount;
        System.out.println("Successfully deposited ₹" + amount);
    }

    public void withdraw(double amount) {
        if (amount <= 0) {
            throw new IllegalArgumentException("Withdrawal amount must be positive");
        }
        if (amount > balance) {
            throw new ArithmeticException("Insufficient funds");
        }
        balance -= amount;
        System.out.println("Successfully withdrawn ₹" + amount);
    }
}
```

```

public class ATM {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        ATM_Machine atm = new ATM_Machine(5000);

        while (true) {
            System.out.println("\n----- ATM MENU -----");
            System.out.println("1. Check Balance");
            System.out.println("2. Deposit");
            System.out.println("3. Withdraw");
            System.out.println("4. Exit");
            System.out.print("Enter your choice: ");

            try {
                int choice = sc.nextInt();
                switch (choice) {
                    case 1:
                        atm.checkBalance();
                        break;

                    case 2:
                        System.out.print("Enter deposit amount: ");
                        double depositAmount = sc.nextDouble();
                        atm.deposit(depositAmount);
                        break;

                    case 3:
                        System.out.print("Enter withdrawal amount: ");
                        double withdrawAmount = sc.nextDouble();
                        atm.withdraw(withdrawAmount);
                        break;

                    case 4:
                        System.out.println("Thank you for using ATM!");
                        return;

                    default:
                        System.out.println("Invalid menu option! Try Again.");
                }
            }

            } catch (ArithmeticException | IllegalArgumentException e) {
                System.out.println("Error: " + e.getMessage());
            }

            } catch (InputMismatchException e) {
                System.out.println("Error: Invalid input. Please enter numbers only.");
                sc.nextLine();
            }

            } finally {

```

```

        System.out.println("Transaction completed.\n");
    }
}
}
}

```

Output:

```

----- ATM MENU -----
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter your choice: 2
Enter deposit amount: 12000
Successfully deposited ₹12000.0
Transaction completed.

```

```

----- ATM MENU -----
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter your choice: 3
Enter withdrawal amount: 4500
Successfully withdrawn ₹4500.0
Transaction completed.

```

```

----- ATM MENU -----
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter your choice: 1
Current Balance: ₹12500.0
Transaction completed.

----- ATM MENU -----
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter your choice: 2
Enter deposit amount: -900
Error: Deposit amount must be positive
Transaction completed.

```

----- ATM MENU -----

1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Enter your choice: 1

Current Balance: ₹12500.0

Transaction completed.

----- ATM MENU -----

1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Enter your choice: 2

Enter deposit amount: -900

Error: Deposit amount must be positive

Transaction completed.

----- ATM MENU -----

1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Enter your choice: 4

Thank you for using ATM!

Transaction completed.

Process finished with exit code 0